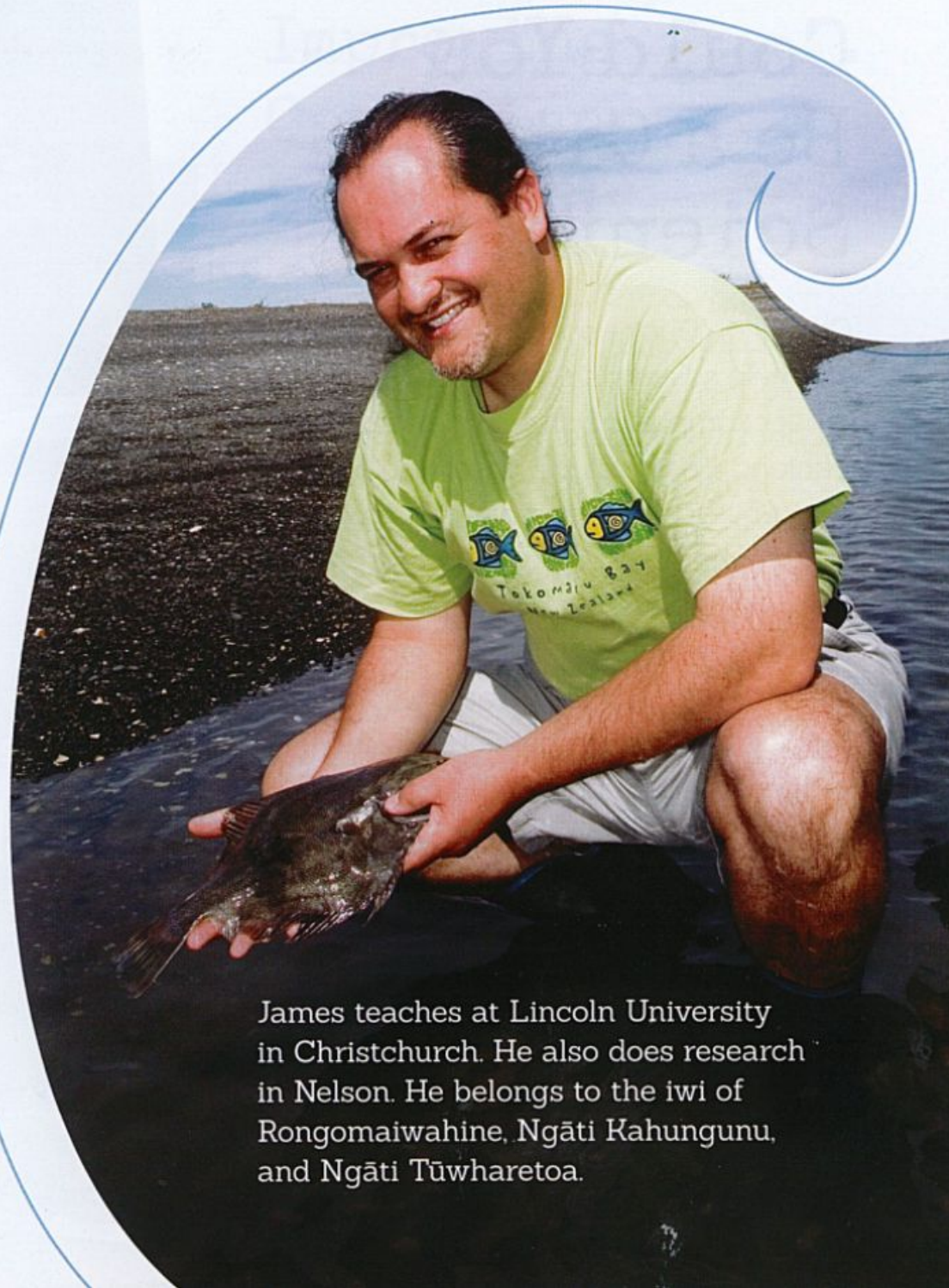


Learning from the Tangata Whenua

An Interview with James Ataria by Susan Paris

*James Ataria is an environmental scientist. He studies how poisonous chemicals affect the natural world. For a long time, he has worried about **pollution** and the problems it's causing. When he was a boy, James's dad took him eeling and to collect kaimoana at the beach. James and his dad spent a lot of time gathering kai from the land as well. Becoming a scientist was a way for James to help protect the things he loves.*

James's work is about studying problems caused by pollution. To do this, he spends a lot of time researching. This isn't unusual. In fact, it's an important part of science. But James works a bit differently from lots of other scientists. In this interview, he tells us more.



James teaches at Lincoln University in Christchurch. He also does research in Nelson. He belongs to the iwi of Rongomaiwahine, Ngāti Kahungunu, and Ngāti Tūwharetoa.

Susan

How do you work differently from other scientists?



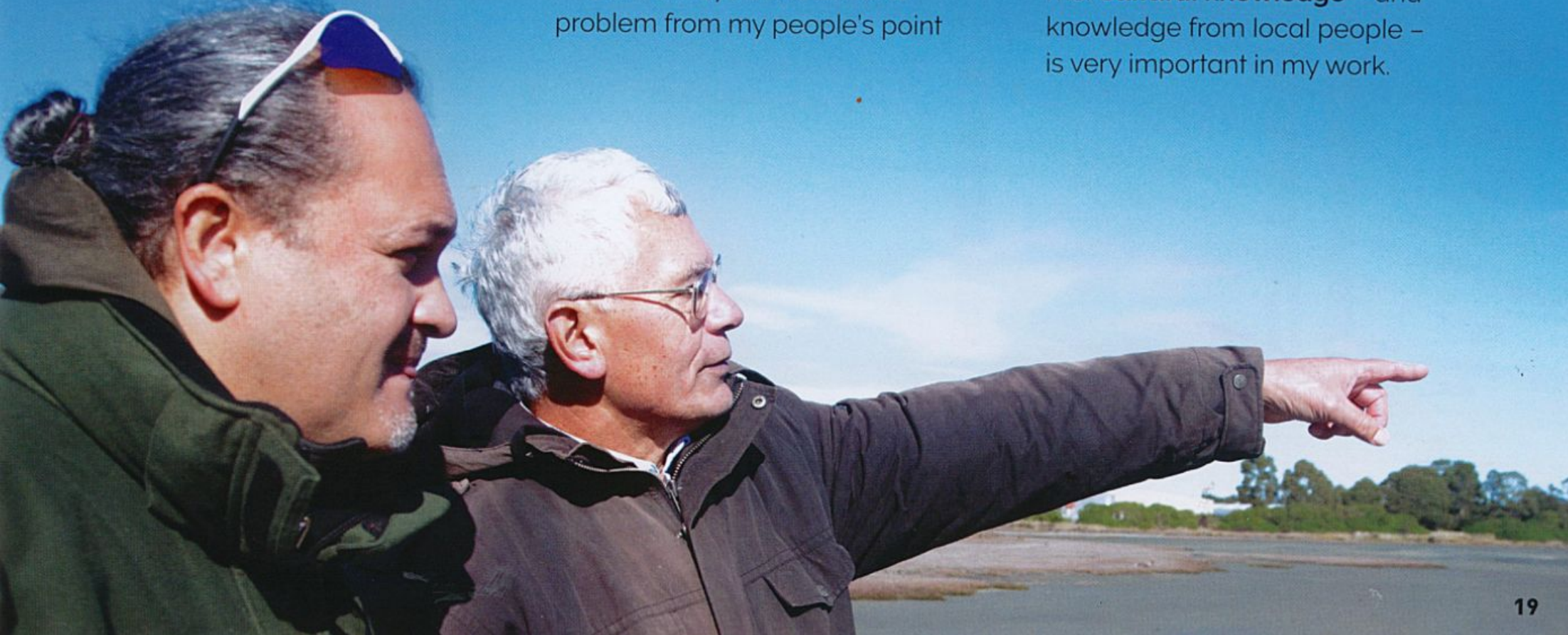
James and another scientist talking about a project

James

I'm a scientist, and I'm also Māori. So I work in two ways. I ask the questions that a scientist asks, and I follow the same process. Science is very specific. Scientists think carefully about how evidence should be collected and **critiqued**. I can't change this – and as a scientist, I wouldn't want to!

But I always think about a problem from my people's point

of view, and I try to get Māori involved with projects from the very beginning. Or at least I try to ask the questions that they would ask. I'm very interested in how my work can benefit Māori, and I spend a lot of time working in Māori communities. But it's a two-way thing: I help these communities, and they help me. **Cultural knowledge** – and knowledge from local people – is very important in my work.



Susan

How does cultural knowledge help you?

James

A few years ago, I worked on a project at the Ahuriri Estuary, near Napier. The estuary, where the river meets the sea, had become very polluted. The **tangata whenua** had put a **rāhui** on taking food from the estuary, because they thought it wasn't safe. Our project was to find out what caused the pollution and how bad it was. One way to gather information was to take a closer look at the **flounder** and **cockles** living in the estuary.



Susan

Who provides the cultural knowledge of this area?

James

Ngāti Kahungunu have lived near the estuary for a very long time. They have watched it change. Scientists are very interested in change. It's often what our research is about. **Kaumātua** from the iwi could tell us things no one else knew, such as when the estuary's flounder and cockles had become harder to find.

Local Māori also told us the best places to catch Ahuriri flounder and the best ways to catch them. This helped my team to plan the way we gathered our **data**. As well, we used the Māori fishing calendar, which told us when we could expect big catches of the **species** we were studying. This cultural knowledge helped our research a great deal. It saved a lot of time.



James at the estuary



Cockles



Susan

Do you trust the cultural and local knowledge?

James

I don't question cultural or local knowledge at all. You can't measure someone's opinion about whether an estuary used to be less polluted. To check something like this, scientists need data. We need to measure things that tell us about the quality of the water. We also keep track of the numbers of species found in the estuary over time. This is evidence that can be checked.

But you can check cultural or local knowledge about a place. For example, you can ask a lot of people the same question. If their answers are the same, then you know it's more likely to be true. As a scientist, you just have to work in the right way with this kind of information.



Collecting samples



Testing samples



Susan

What is the best way to find out about cultural knowledge?

James

To find out about Māori cultural knowledge, you need to embrace these communities and their culture. When I'm working with a Māori community, I always try to stay in that community. If there are things that we need to participate in, like **pōwhiri** and **poroporoaki**, then I make sure everyone on the team goes. These cultural events are important. We get to meet the tangata whenua, and talk with them. We can learn about the

importance of the local environment to them – and what it means if that environment is now polluted.

It's important that scientists listen to people. It encourages us to think more about our work and how we can help others. It also makes us think about the best way to work with a community. That includes both receiving knowledge and giving knowledge back. It's always good to have an open mind, especially when you're a scientist.



Members of the local community at the Ahuriri Estuary





Susan

Why do you say that it's good for scientists to be open-minded?

James

Scientists test ideas and answer questions. But some things can't be explained by science. And sometimes, we get results that surprise us.

Some results can be difficult to figure out or explain. Having an open mind means that you're more likely to think about a problem in a new way. Then you'll ask different questions, ones that you haven't thought of before. Maybe you'll find the answer that way.

I've learned one important thing in my work: there are not many things in this world we can be one hundred percent certain about. This is especially true in science.



Scientific Glossary

cockles – a type of shellfish

critiqued – carefully analysed and questioned

cultural knowledge – things that people understand or know from their background

data – information such as measurements and observations

flounder – a type of flatfish that often lives in estuaries

pollution – dirty and unhealthy air, water, or soil

species – living things that share similar characteristics and may look the same

Māori Glossary

kaumātua – respected elders

poroporoaki – farewell

pōwhiri – welcome

rāhui – a ban on fishing and collecting kaimoana

tangata whenua – local people; people who belong to the land in a tribal area